

Handwritten Arabic Calligraphy-based CAPTCHA

Master's Thesis

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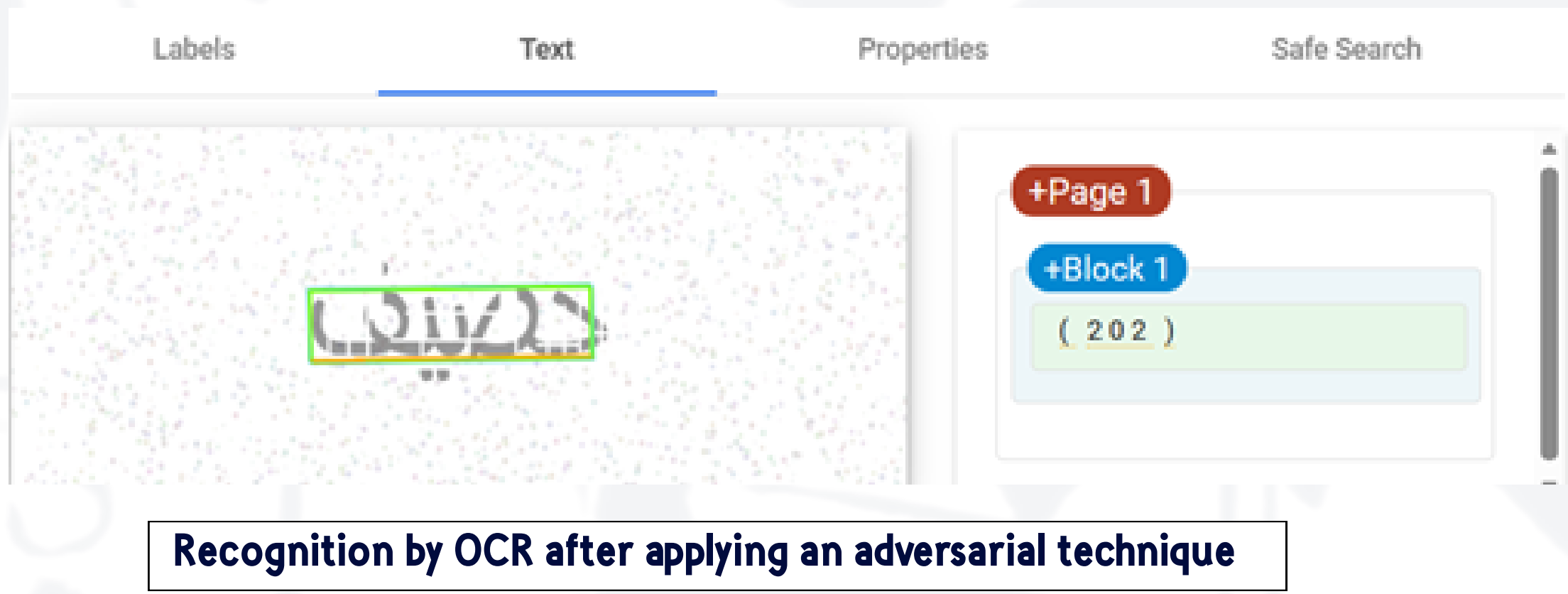
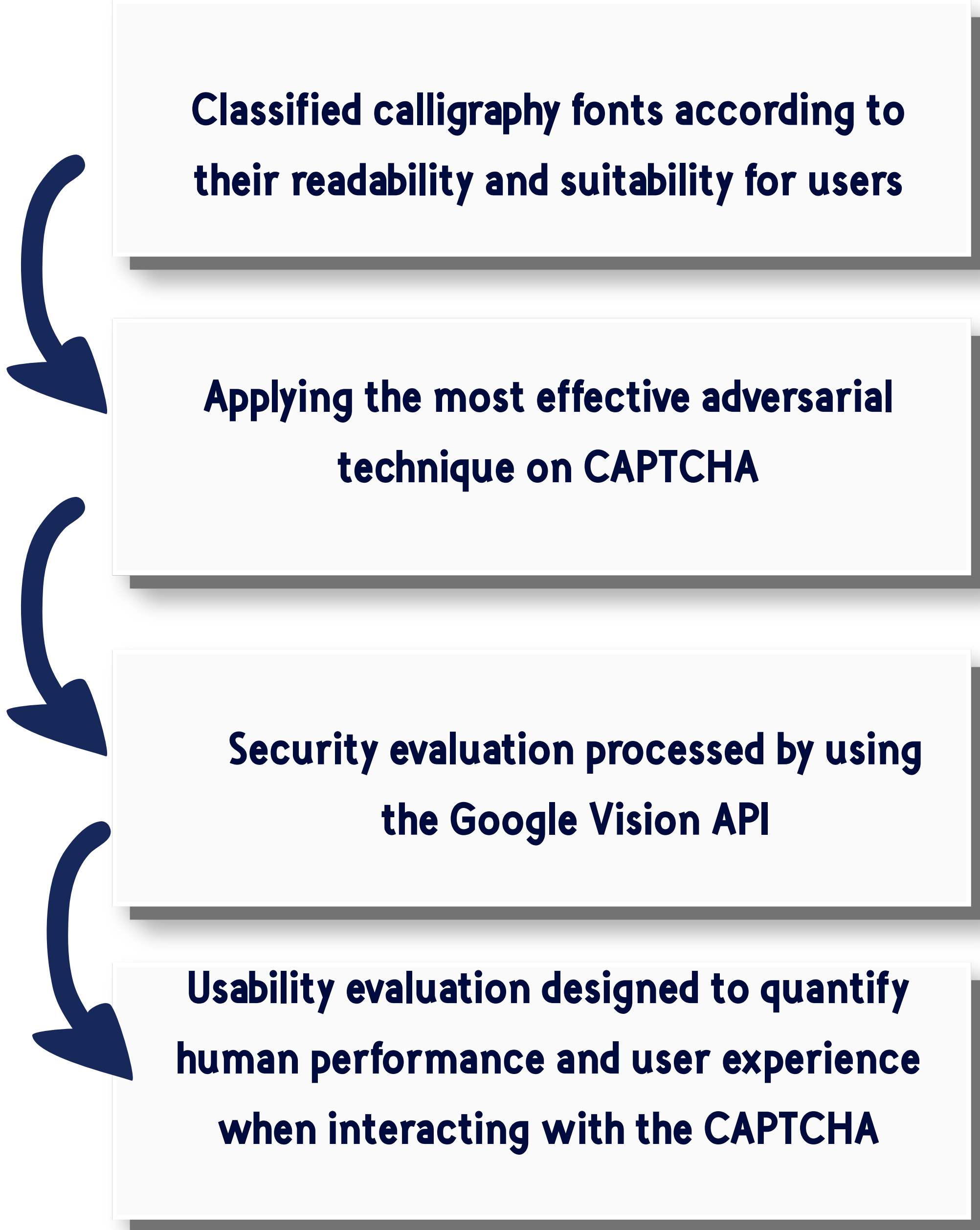
INTRODUCTION

CAPTCHA are widely used to distinguish human users from automated attacks. With the increasing number of Arabic internet users, Arabic-based CAPTCHAs have become increasingly important. This study proposes generating handwritten Arabic CAPTCHAs using diverse calligraphic styles and adversarial techniques. It evaluates the trade-off between security and usability to assess their effectiveness against automated attacks.

METHODOLOGY

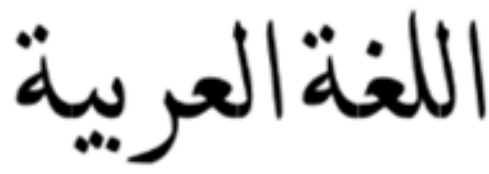
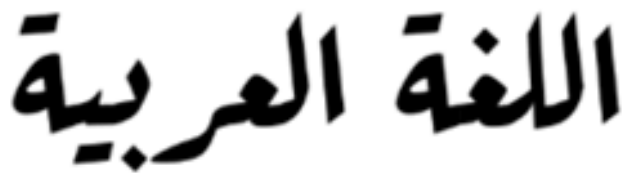
- 1 Studying literature text-based CAPTCHAs.
- 2 CAPTCHA samples generation
- 3 Font classification model
- 4 Apply the Adversarial Techniques
- 5 Evaluating security and usability

EXPERIMENT



CONCLUSION

This study proposes a framework for generating and evaluating handwritten Arabic CAPTCHAs using calligraphic styles and adversarial techniques. The results show that both style and text meaning significantly impact security and usability, highlighting the effectiveness of calligraphic variation in improving CAPTCHA robustness.

 (a) Thuluth	 (b) Naskh
 (c) Diwani	 (d) Ruqaa

Examples of Arabic calligraphic styles.